

# The Taylor tree for absolutely summable coefficient families (Taylor-tree Stage 4)

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## Abstract

Units 241–247 complete Stage 4 of the Taylor-tree programme (Astra #28): the polynomial theorem of Stage 3 is extended to phase and amplitude given on the unit box by *absolutely summable power series*  $\xi(u) = \sum_{\gamma} c_{\gamma}^{\xi} u^{\gamma}$ ,  $\eta(u) = \sum_{\gamma} c_{\gamma}^{\eta} u^{\gamma}$  with  $\sum_{\gamma} |c_{\gamma}| < \infty$  — the hypothesis implied by the paper’s holomorphy on a polydisc  $\bar{D}_R$ ,  $R > b = 1$ , through Cauchy estimates. For  $\beta > 0$ ,  $k_i > 0$ , every cutoff  $L > 0$  and every  $N \geq 1$ ,

$$\left| Z(N) - \sum_{\mu \in \Lambda_L} N^{-\mu} \sum_{j \leq d-1} A_{\mu,j} (\log N)^j \right| \leq C(\xi(0), \|c^{\eta}\|_1, \|c^{\xi}\|_1) N^{-L} (1 + \log N)^{d-1}$$

(Headline XXVII, `familyTaylorTree_cutoff_bound`), with cutoff-free coefficients  $A_{\mu,j}$  vanishing off  $\Lambda(h, k)$  and a constant depending on the data only through  $\xi(0)$  and the two coefficient masses; hence  $O(N^{-L}(\log N)^{d-1})$  and  $o(N^{-L'})$  for  $L' < L$ . The route is *completion by density*, not a redo of Stage 3 for infinite families: box truncations are genuine monomial lists, the truncated coefficients form a Cauchy sequence by a Lipschitz *stability gate*, the truncated integrals converge by dominated convergence at fixed  $N$ , and the uniform polynomial bound passes to the limit. Numerics ( $d = 1$ ,  $\xi = 0.3e^{u/2}$ ,  $\eta = (1 - u/3)^{-1}$ ,  $L = 5/2$ ):  $\text{error} \times N^{5/2} \approx 0.05\text{--}0.06$  for  $N = 10, \dots, 640$ . No `sorry` and no additional `axiom` declarations; 9123 jobs.

## 1 Architecture

**List algebra (u241, MonoRepPerm).** The coefficient functionals of Stage 3 depend on a monomial list only through its multiset of  $(\gamma, c)$  pairs: invariant under permutation, additive under `++`; the list product distributes over `++` (exactly on the left, up to permutation on the right). The **power-difference estimate**  $\text{pow}(J ++ \Delta)^p \sim \text{pow}(J)^p ++ R_p$  with  $\|R_p\|_1 \leq p\|\Delta\|_1(\|J\|_1 + \|\Delta\|_1)^{p-1}$  is the list form of  $\text{mass}(x^p - y^p) \leq pB^{p-1}\text{mass}(x - y)$ .

**Stability gate (u242, CoeffStability).** If  $\eta' \sim \eta ++ \Delta\eta$ ,  $J' \sim J ++ \Delta$ ,  $\xi'(0) = \xi(0) = a$ ,  $\|\eta\|_1 \leq E$ ,  $\|J\|_1 + \|\Delta\|_1 \leq B$ , then for  $\mu > 0$

$$|A_{\mu,j}(\xi', \eta') - A_{\mu,j}(\xi, \eta)| \leq K_k D(E\beta M_{\mu+1/2,n}(a+B)\|\Delta\|_1 + M_{\mu,n}(a+B)\|\Delta\eta\|_1),$$

where the factor  $p$  from the power difference is absorbed by shifting the phase order,  $M_{\nu,n,p+1} = M_{\nu+1/2,n,p}$ , and the phase orders are summed by the Tonelli identity of unit 234. Astra #28 named this the single riskiest step of Stage 4; it passed in one unit.

**Coefficient families (u243, CoeffFamily).** Families  $c : \mathbb{N}^d \rightarrow \mathbb{R}$  with  $\sum |c_\gamma| < \infty$ ; mass, evaluation ( $|\text{eval } c u| \leq \text{mass } c$  on the cube), box truncations  $\{\gamma : \gamma_i \leq m\}$  as monomial lists keeping the constant term; the tail mass  $\sum_{\gamma \notin \text{box}_m} |c_\gamma| \rightarrow 0$  (Tannery); successive truncations differ by an appended list of mass at most the tail mass; uniform convergence on the cube.

**Integral convergence (u244).**  $Z_m(N) \rightarrow Z(N)$  at fixed  $N \geq 0$  by dominated convergence with the constant bound  $\text{mass}(\eta) e^{\beta \sqrt{N} \text{mass}(\xi)}$  on the box.

**Coefficient convergence (u245).**  $|A(m') - A(m)| \leq C_1 \text{tail}_\xi(m) + C_2 \text{tail}_\eta(m)$  for  $m \leq m'$ , so the truncated coefficients are Cauchy;  $A_{\mu,j}(c^\xi, c^\eta)$  is their limit (at  $\mu \leq 0$  and off  $\Lambda(h, k)$  all terms vanish, so the limit does).

**Uniform constants (u246).**  $M_{\nu,r}(b)$  is increasing in  $b$  and the tail constant in  $|b|$ , so the polynomial cutoff constant is bounded by an explicit function of  $(\xi(0), E, B)$  whenever  $\|\eta\|_1 \leq E$ ,  $\|J\|_1 \leq B$ .

**Assembly (u247).** Fix  $N \geq 1$ ; apply the uniform polynomial bound to every truncation and let  $m \rightarrow \infty$  on both sides (`le_of_tendsto`). No exchange of the asymptotic limit with the truncation limit is needed.

## 2 Correspondence with the paper

Lean's  $N$  is the sample size  $n$ ; the dimension is  $d = n_{\text{Lean}} + 1$ ;  $(\log N)^j$  carries the paper's  $m = j + 1$ . The hypothesis is the coefficient-summability form of the analytic hypothesis of `thm:TaylorTree`; the Cauchy-estimate bridge from holomorphy on  $D_R$  to  $\sum |c_\gamma| < \infty$  is a separate (unformalised) obligation, as is  $b \neq 1$  (scaling  $Z_b(N) = b^{|h|+d} Z_1(N b^{2|k|}; \xi(b \cdot), \eta(b \cdot))$ ) and the derivative dictionary  $\beta^p \int t^{\mu-1} (-\log t)^i (\sqrt{t})^p e^{-\beta t + \beta a \sqrt{t}} = (-\partial_\mu)^i \partial_a^p S_\mu(a)$ .

## 3 Lessons

Complete by density: keep Stage 3 for lists, extend only the coefficient algebra and functionals. Avoid the infinite Cauchy product (`Fin d`  $\rightarrow \mathbb{N}$  has no `HasAntidiagonal` instance): truncation differences are literal appended lists, so permutation invariance replaces convolution algebra. Lean: `tendsto_tsum_of_dominated_convergence` lives in `Mathlib.Analysis.Normed.Group.Tannery` (import it explicitly); use `List.perm_of_nodup_nodup_toFinset_eq` with `List.mem_toFinset` rather than `Finset.toList_toFinset`; `limUnder + cauchySeq_tendsto_of_complete` for a limit defined without choosing a representative; `cauchySeq_of_le_tendsto_0` wants a bound at the smaller index, so tail masses must be shown antitone.